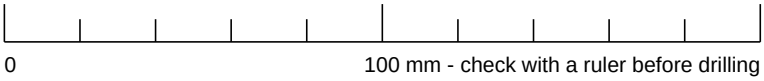
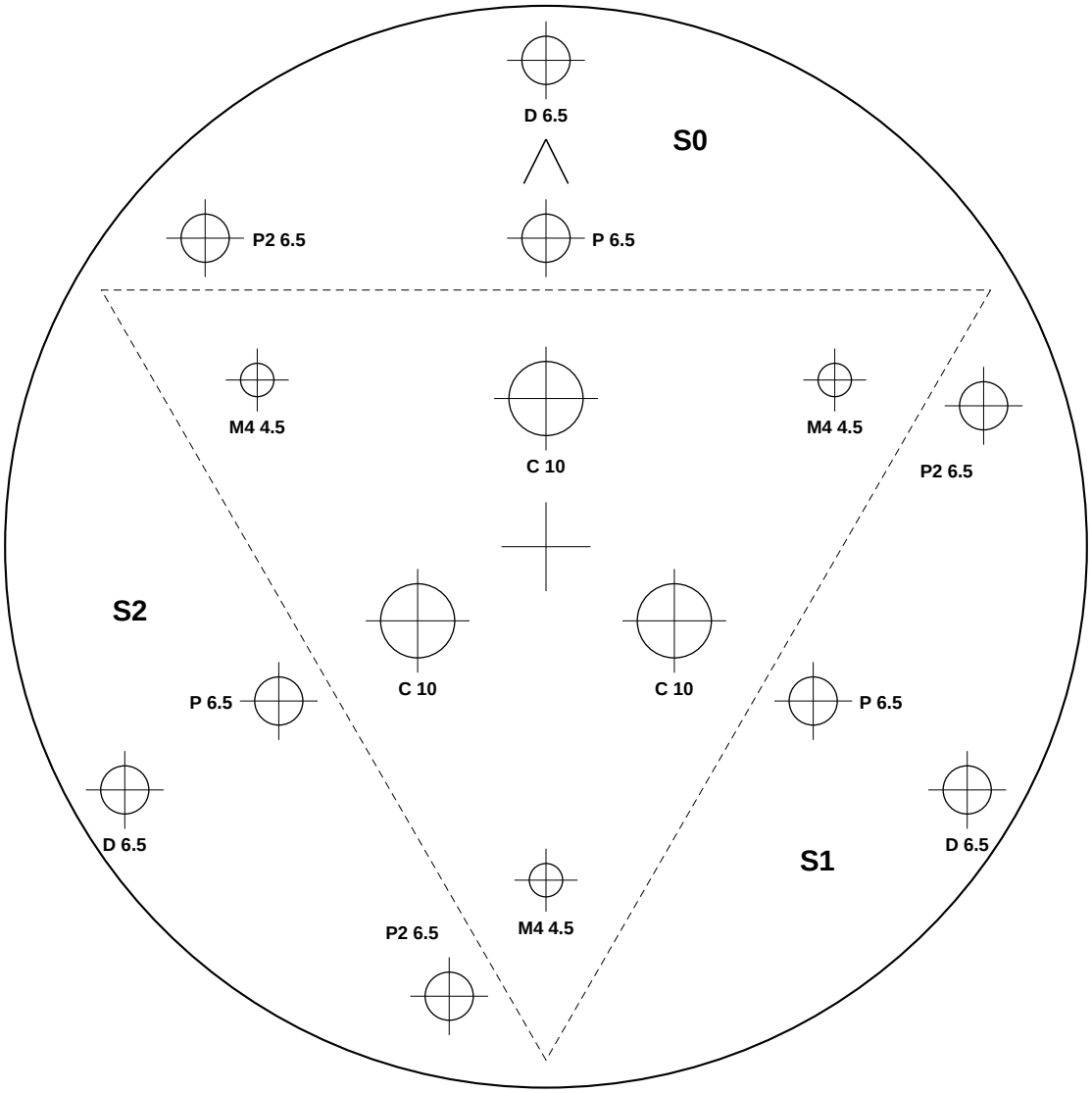
# Shield deck drill template - 146 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP (electronics side). Tape to the board, centre-punch each crosshair, then drill.



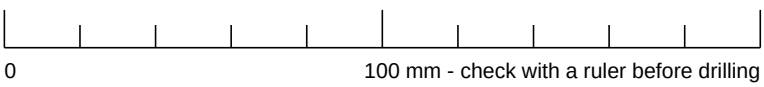




Solid circle = cut line (146 mm). Dashed triangle = where the 120 mm back panels stand, copper side out.

D 6.5 mm = rubber duck / monopole (31 mm in front of face) P / P2 6.5 mm = patch (7 mm in front of face; P2 14 mm from the face end)

C 10 mm = dipole coax pass-through (inside core) M4 4.5 mm = bolts to the electronics plate (inside core)

Arrow points to Sector 0. Holes are symmetric, so the template also works flipped (labels read mirrored).

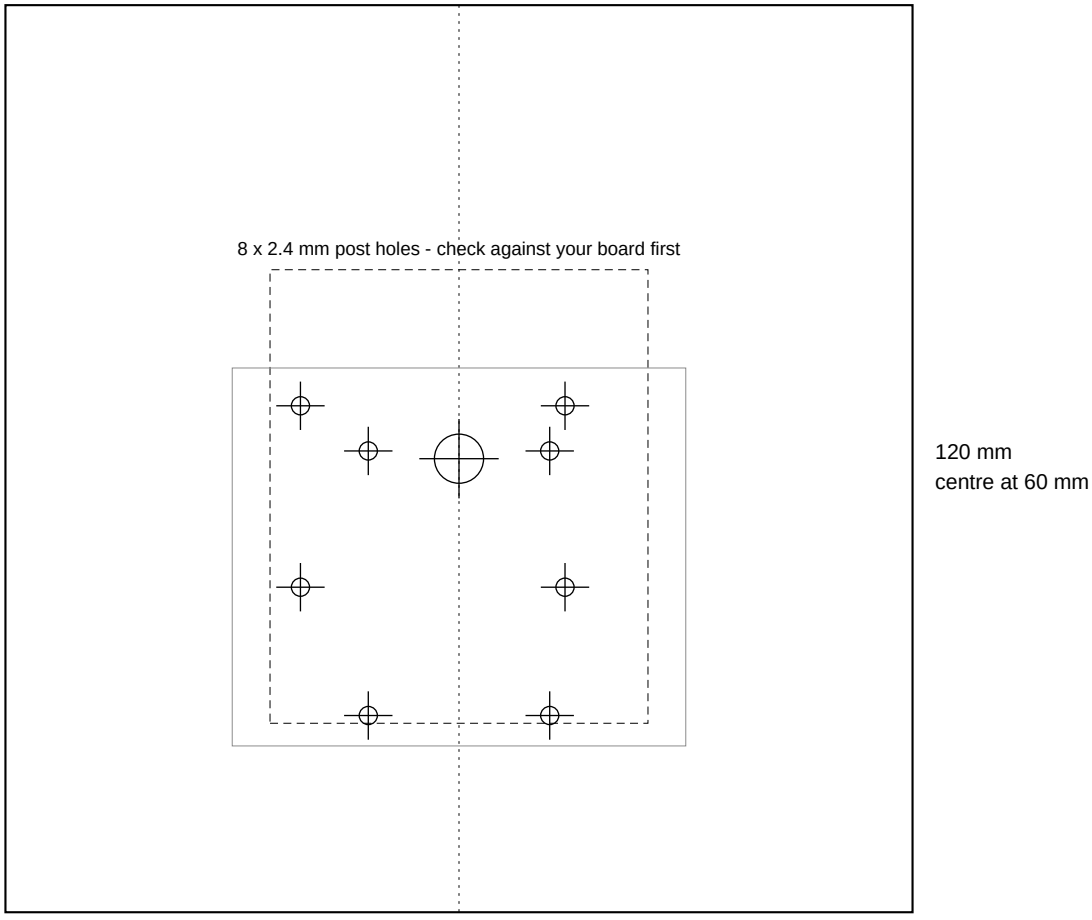


# Back panel (reflector face) template - 120 x 120 mm - PRINT AT 100%

Make 3. Viewed from the FRONT (copper side, facing the antenna). Top edge goes against the shield deck.

0 100 mm - check with a ruler before drilling

**TOP - against shield deck**




A (dashed): bar vertical, notch right, board 35-95 mm down - stalk reaches the P2 deck jack

C (grey): bar horizontal, notch up, board 48-98 mm down

Centre 6.5 mm hole: DIY dipole RP-SMA only

Rubber duck / monopole / dipole sit 31 mm in front of this face (on the D hole).

Panel board 50 x 60 mm: 7 mm off this face on nylon standoffs. A = stalk on the P2 deck jack; C = coax may need extending.

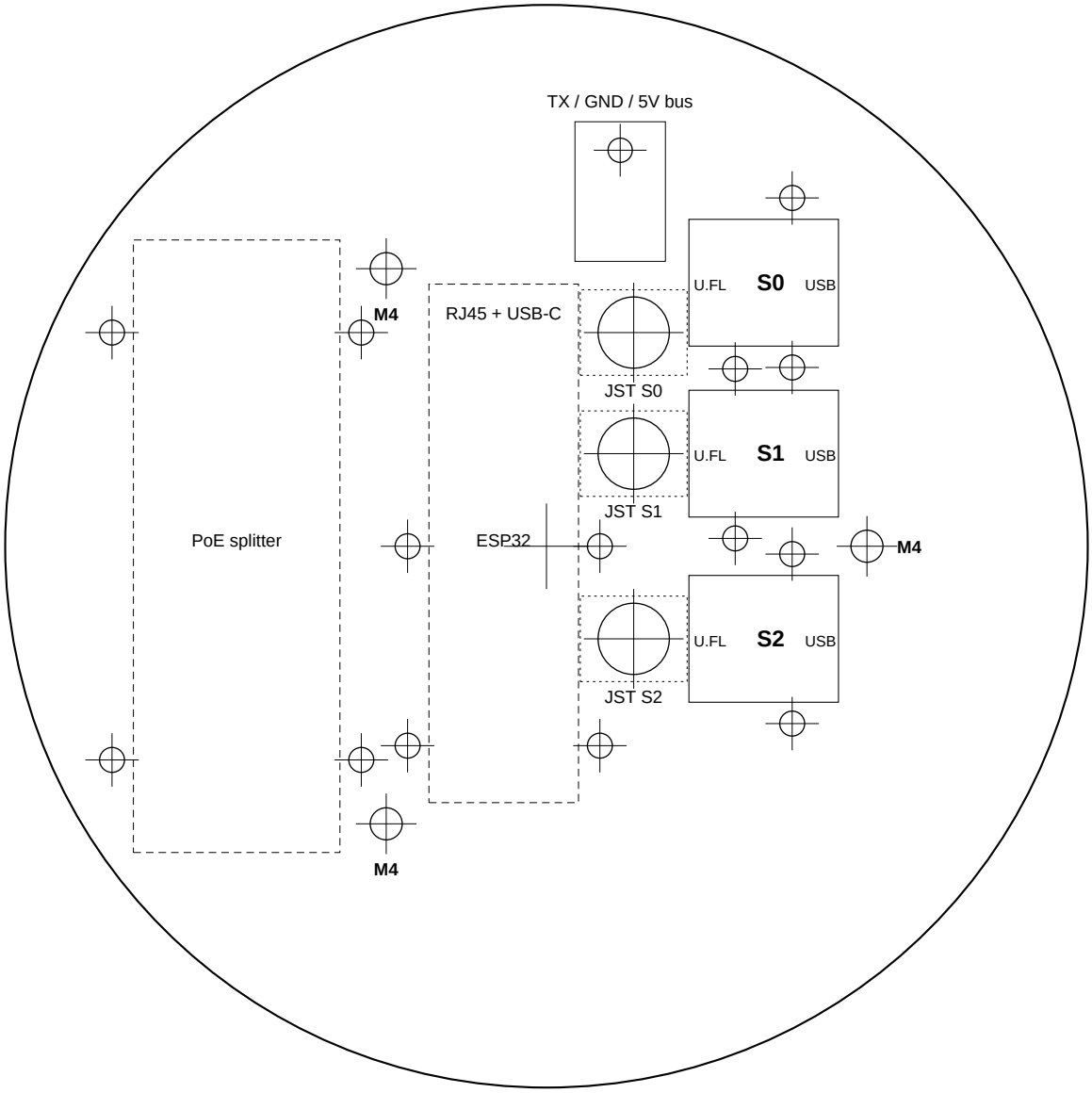
Corners meet at 120 deg to form the triangle; solder or tape the seams. Holes under 10 mm don't affect the reflector.

0 100 mm - check with a ruler before drilling

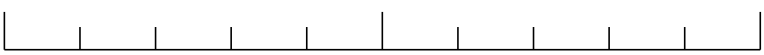
# Electronics plate template - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP (parts side). Gateway left, scanners right, pigtails between, each JST pair over its pigtail hole.

0 100 mm - check with a ruler before drilling



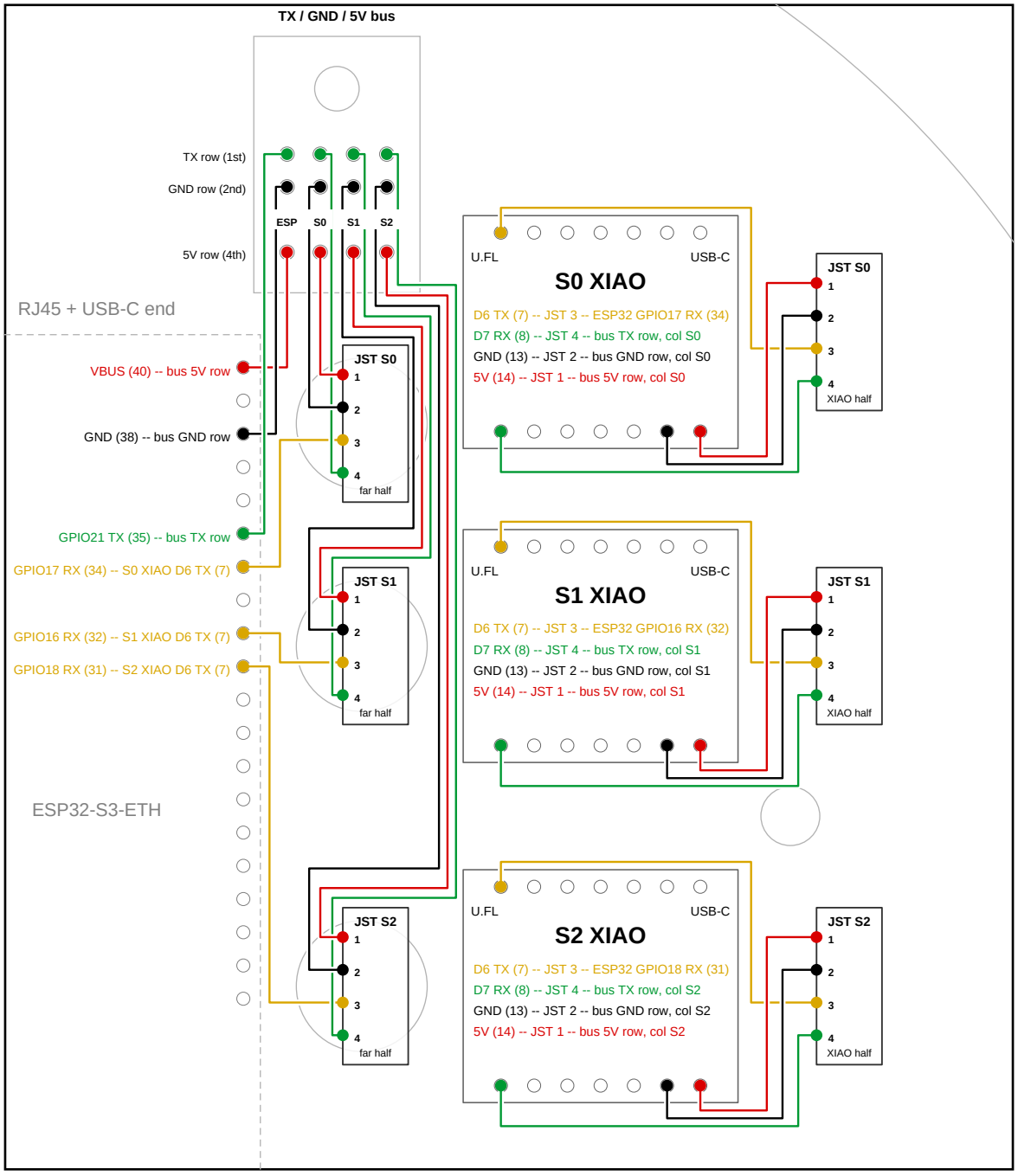

3 x 10 mm pigtail holes beside the XIAOs. 3.4 mm (M3): bus board. 3 x 4.5 mm (M4, labelled): bolts to the deck.
Line up the M4 holes with the deck's, plate 12 o'clock 30 deg counter-clockwise of the deck's S0 arrow.
Small holes: 3.5 mm zip-tie pairs, one tie across each XIAO (USB-C toward the rim), two each across splitter and ESP32.
Dotted: mated JST pair over each pigtail hole (unplug to reflash); far side D6 to its ESP32 RX pin, 3 wires to the bus.
Dashed outlines: ESP32-S3-ETH 72.8 x 21 mm; splitter size estimated from a photo - lay it on the
printout and move its tie holes to 3 mm outside the real edges before drilling.



0 100 mm - check with a ruler before drilling

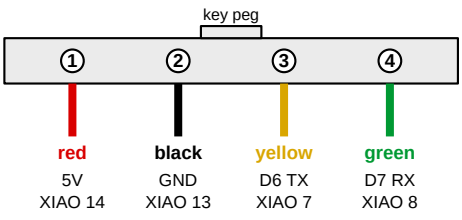
# Electronics plate wiring - same layout as the drill template

Top view of the radio side, 2x (not for drilling). Pin numbers in brackets. Pad positions approximate.
JST halves drawn apart: XIAO half right of each radio, far half on its pigtail hole; same name = they plug together.




D6 -> its own ESP RX (the pin is the sector): S0 GPIO17 (34), S1 GPIO16 (32), S2 GPIO18 (31).
ESP TX GPIO21 (35) -> bus TX row -> every D7. Feed: TX (35) -> TX row, VBUS (40) -> 5V row, GND (38) -> GND row.
Bus: hole end up, rows bridged, 3rd row spare. ESP32 RJ45 side up, RJ45 / USB-C end north: VBUS (40) top right pin.

**JST head (either half), crimp side, key peg up:**




Pins left to right: 1 red 5V, 2 black GND, 3 yellow D6 TX, 4 green D7 RX.
Male and female halves carry the same colours in the same order on the crimp side, so the XIAO tail and the far tail match colour for colour through the joint.
XIAO tail: red -> 5V (14), black -> GND (13), yellow -> D6 (7), green -> D7 (8).
Far tail: red -> bus 5V row, black -> bus GND row, green -> bus TX row, yellow -> that sector's ESP RX pin (labels on the drawing). Female half on the far tail keeps 5V covered.